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PETROLEUM TRAINING INSTITUTE, EFFURUN

CTE 112 — Past Questions & Materials

by Renance

Course: CTE 112

Institution: Petroleum Training Institute, Effurun

Papers in this pack: 1

Levels: ND1

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Paper 1 of 1 — CTE 112 ELECTRICAL WORKSHOP PRACTICE AND TECHNOLOGY MODULE 1

Level ND1 • First Semester • Handout

ELECTRICAL WORKSHOP PRACTICE AND TECHNOLOGY (CTE 112)

Module 1

By

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2022 March 27 ALKANCHI

Week IV

ELECTRICAL HAZARDS

- A serious workplace hazard that exposes workers to burns, electrocution, shock, arc flash/arc blast, fire, or explosion.

- By Identifying these hazards and understanding

Full Name how they happen,

2022 March 27 we

Week IV

can take steps to protect ourselves.

Importance of Electrical Safety

- A set of guidelines personnel follow to mitigate electrical hazards and prevent its dangerous effects in case of an incident.

- It is important to take proper electrical safety measures to help avoid any type of accident or injury,

Full Name especially 2022

when

March 27 it comes to high voltage

Week IV

machinery.

Electricity and Electric Circuit

- Electricity is simply the flow of electric charge. It provides energy or power that is used to run electrical and electronic appliances. Electric current is the rate of flow of charges rated in Amps

- FullElectric

Name circuit is2022

theMarch

closed

27 path provided for the

Week IV

flow of Electric charges.

Electric shock

- Occurs when part of your

body becomes part of the

electric circuit. Severity

depends on;

- Path of current through the body

- Amount of current through the

body

Full Name 2022 March 27

- Amount of time the body is

Week IV

in the

circuit.

How to help an electric shock victim

- Turn off the power or use a non-

conductive object, such as a

wooden broom handle or a

rubber object, to push the victim

away from the electrical source.

- FullCall

Name for emergency2022medical

March 27

Week IV

services.

How to help an electric shock victim cont'd

- If the victim has burns, cool the affected area with

cool water for at least 10-15 minutes to help

reduce pain and swelling Do not yet ice to avoid

skin damage.

- Keep the victim warm and comfortable until

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medical help arrives.

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Precautions against electric shock

- Turn off power before starting maintenance work.
- Test for power in the circuit after turning off the circuit breaker for confirmation.
- Use insulated tools for electrical work.
- Safely discharge capacitors before working on capacitive circuits.

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Week IV

Basic rules of electrical safety

- Always use caution when working with electricity
- Always unplug electrical appliances before attempting to repair or modify them.
- Never touch an electrical appliance or electrical outlet with wet hands.
- Always use a properly grounded outlet when

Full Name 2022 March 27

plugging in an electrical

Week IVappliance.

Basic rules of electrical safety cont'd

- Keep electrical appliances and cords away from water.
- Use caution when handling electrical cords
- Do not touch live electrical wires or terminals
- If you are unsure about how to safely use an electrical appliance or perform electrical work, consult a qualified electrician

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. Week IV

Who is a qualified person ?

- A qualified electrical person is an individual who has the knowledge, skills, and experience necessary to safely perform electrical work.
- He has a valid electrical license or certification, which is issued by the appropriate regulatory agency after he has completed

Full Name the necessary

2022training

March 27 and demonstrated their skills

Week IV

through testing.

ELECTRICAL SAFETY EQUIPMENT

- Electrical safety equipment refers to the tools and devices that are used to protect individuals from electrical hazards and to ensure the safe operation of electrical systems and equipment.

- Examples of electrical safety equipment are, electrical

Full Name gloves, electrical mats,

2022 March 27 voltage detectors,

Week IV

circuit testers, arc flash suits, etc.

Personal Protective Equipment

- Personal protective equipment, or PPE, refers to any type of clothing or equipment that is worn to protect the wearer from hazards in the workplace. PPE can include items such as hard hats, safety glasses, gloves, respirators, earplugs, and other protective clothing.

- The purpose of PPE

Full Name is to

2022 March 27 reduce the risk of

Week IV

injury or illness to workers

Body protection

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Week IV

Coverall and jacket

Head protection

Full Name 2022 March 27

Hard hat

Week IV

Eye protection

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Eye glasses

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Face protection

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Face shield

Week IV

Hand protection

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Hand Gloves

Week IV

PPE for electrical works

- Cotton protective clothing with long sleeves
- Helmet or hard hat
- Goggles for eye protection
- Gloves (leather or rubber)
- Hearing protectors
- FullSafety

Name footwear 2022 March 27

Week IV

First aid procedure for the eyes (particle)

- Stay calm and reassure the person who was injured.
- Have the person gently blink their eye to try and remove the object.
- If the object does not come out, do not try to remove it with your fingers. Instead, use the corner of a clean, soft cloth to try and remove it.
- Full

If Name

the object cannot be removed

2022 March 27

or if the eye is red

or painful, seek medicalWeekattention

IV immediately.

First aid procedure for the eyes (chemical)

- Stay calm and reassure the person who was injured.
- Have the person immediately flush their eye with water for at least 15 minutes. Water should be at room temperature
- If the chemical is a powder, have the person blow their nose to remove any remaining particles.
- IfFullthe

Name eye is red or painful, seek medical attention

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immediately.

Lock-out devices

- A lock-out device is a safety device that is used to prevent the unintended start-up or activation of machinery or equipment. It is typically used in industrial settings to protect workers from accidental injury when servicing or repairing machinery.

- Types of lock-out devices include padlocks, hasps,

Full Name 2022 March 27

lockable gate valves, lockable electrical circuit

Week IV

breakers, and lockable disconnect switches.

Lock-out devices

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Week IV

ELECTRICAL SAFETY PROCEDURE

- Electrical safety procedures are a set of guidelines and rules that are put in place to protect individuals from the hazards associated with electrical equipment and systems.

- These procedures are designed to help prevent accidents

Full Name and injuries 2022 that

March can

27 occur as a result of

Week IV

electrical shocks, fires, and other electrical hazards.

Symptoms of an electric shock victim

- Burns
- Pains
- Numbness or tingling
- Weakness or fatigue
- Confusion or disorientation
- Difficulty breathing

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Fire extinguisher classification

- For a fire involving flammable liquids, such as gasoline, oil, or paint, it is recommended to use a Class B fire extinguisher. These extinguishers are designed to extinguish

fires involving flammable liquids and gases by interrupting the chemical reaction that is fueling the fire. The most common types of Class B extinguishers are foam, CO2, and dry chemical.

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Fire extinguisher classification cont'd

- For a fire involving energized electrical equipment, it is recommended to use a Class C fire extinguisher. These extinguishers are designed to extinguish fires involving electrical equipment, such as appliances, computers, and wiring, by interrupting the flow of electricity. The most common type of Class C extinguisher is a CO2 extinguisher, which is safe to use on electrical fires because it does not

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conduct electricity. Week IV

FUSES AND CIRCUIT

BREAKERS

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Week IV

FUSE

- A fuse is a type of electrical safety device that is used to protect circuits and equipment from excessive current. It consists of a thin strip of conductive material, usually made of metal, that is connected in series with the circuit

2022 March 27

or equipment. Week IV

Dual element cartridge fuse

- A dual element cartridge fuse is a type of electrical fuse that is designed to protect against both overload and short circuit conditions. It consists of two elements, or fusible links, that are connected in parallel within a

2022 March 27

single cartridge. Week IV

Working of a dual element cartridge fuse

- The first element is designed to protect against overloads, and is made of a fusible alloy that is sized to melt at a specific current level. If the current in the circuit exceeds the rated level of the fuse, the fusible alloy will melt, interrupting the flow of electricity and protecting the circuit from damage.
- The second element is designed to protect against short circuits, and is made of a conductive material that is designed to melt at a much lower current level than the first element. If a short circuit occurs, the current in the circuit will increase dramatically, causing the second element to melt

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and interrupt the flow of electricity.

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NEC rules for installing fuses

- Fuses must be properly sized for the circuit they are protecting. The NEC provides tables with recommended fuse sizes based on the size of the conductor and the expected load on the circuit.
- Fuses must be installed in an approved enclosure or fuseholder. The NEC specifies the minimum requirements for these enclosures, including the

2022 March 27

materials they must be

made

IV

of and the types of fuses

that are allowed to be used with them.

NEC rules for installing fuses cont'd

- Fuses must be located in an easily accessible location. The NEC requires that fuses be located within sight of the equipment they are protecting and within reach of the operator.

- Fuses must be properly marked. The NEC requires that fuses be marked with their rating and the type of circuit they are

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intended to protect. Week IV

CIRCUIT BREAKER

- A circuit breaker is a device that is designed to interrupt the flow of electricity in an electrical circuit when it becomes overloaded.

Circuit breakers are used to protect electrical circuits and devices from damage caused by excess current.

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Molded case circuit breaker

- A molded case circuit breaker (MCCB) is a type of electrical protection device that is used to protect an electrical circuit from damage caused by overload or short circuit. It is designed to automatically detect and interrupt the flow of electrical current when it exceeds a predetermined level, thereby protecting the circuit from damage.

- They are commonly used in residential, commercial, and industrial electrical systems to protect against electrical hazards and ensure the safety of electrical equipment and personnel. MCCBs can be manually operated or automatically

tripped by

March 27

the trip unit, which is designed to detect and respond to overcurrent

Week IV conditions.

Molded case circuit breaker

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Week IV

Fuse replacement steps

- Identify the type of fuse you need to replace.
- Locate the fuse panel or box.
- Turn off the power.
- Open the fuse panel or box.
- Identify the faulty fuse.
- Remove the faulty fuse.
- Install the new fuse.
- Close the fuse panel or box and turn the power back on.

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Circuit breaker reset steps

- Locate the circuit breaker panel. This is usually found in a central location in the home, such as a utility room, basement, or closet.

The panel may also be located near the electrical meter or outside the home near the main electrical service panel.

- Turn off any appliances or electronics that are plugged in. This

will help to prevent any potential damage to the appliance or electronics if the circuit breaker trips again.

- Identify the tripped circuit breaker. A tripped circuit breaker will typically be in the "off" position or have a switch that is not fully

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in the "on" position. You may also see a red or black switch or

Week IV

handle on the circuit breaker.

Circuit breaker reset steps cont'd

- Reset the circuit breaker. To reset the circuit breaker, you'll need to switch it back to the "on" position. This can usually be done by simply flipping the switch or handle.

- Turn the appliances or electronics back on. Once the circuit breaker has been reset, you can turn the appliances or electronics back on to test if the issue has been resolved.

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THANK YOU

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